

ROSEVELLE

Cosmetic Product Recommendation and Retail Analytics System

*A Major Project Report Submitted in Partial Fulfillment of the Requirements
for the Degree of Bachelor of Technology in Computer Science & Engineering*

Submitted By:

[Student Name / Registration No.]

Under the Guidance of:

[Project Guide Name & Designation]
Department of Computer Science & Engineering

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
FACULTY OF ENGINEERING & TECHNOLOGY
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BONAFIDE CERTIFICATE

This is to certify that the project report titled "**ROSEVELLE: Cosmetic Product Recommendation and Retail Analytics System**" is a bonafide work carried out by [Student Name] (Registration No. [XXXXXXXX]) under my supervision and guidance in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering.

To the best of my knowledge, the matter embodied in this project report has not been submitted to any other University or Institution for the award of any degree or diploma.

Project Guide

Department of CSE

Head of Department

Department of CSE

ABSTRACT

In modern e-commerce systems, providing personalized recommendations while simultaneously extracting actionable business intelligence from retail transaction data remains a critical operational challenge. The Rosevelle project presents an end-to-end e-commerce recommendation and retail analytics software platform specifically engineered for the luxury cosmetics domain.

The system addresses two complementary operational domains through a strict dual-dataset architecture: (1) Personalized product recommendation powered by real-world consumer ratings from the Amazon All Beauty dataset (comprising 85 validated review triples across 15 active users and 12 flagship cosmetic products), and (2) Market basket analysis and customer segmentation powered by a realistic synthetic e-commerce transaction dataset (121 orders across 30 customer profiles, generating \$9,68,799.00 total revenue).

To achieve high predictive accuracy, Rosevelle implements a multi-algorithm machine learning framework comprising Matrix Factorization via Truncated Singular Value Decomposition (SVD, $k=4$), User-Based Collaborative Filtering, Item-Based Collaborative Filtering, and Content-Based Filtering using TF-IDF metadata vectorization. A hybrid recommendation engine linearly blends SVD ratings with Content-Based TF-IDF cosine similarity scores ($\alpha=0.6$), achieving an empirical Root Mean Squared Error (RMSE) of 1.0751 and Mean Absolute Error (MAE) of 0.8739 across benchmark evaluation splits.

For retail analytics, Rosevelle integrates Apriori Association Rule Mining (discovering 14 co-purchase rules with up to 66.7% confidence and 1.85x lift), RFM customer segmentation via K-Means clustering ($K=4$, Silhouette Score = 0.482), and Multidimensional Sales OLAP aggregation cubes. Automated software testing confirmed 100% verification across 12 API and UI test suites.

CHAPTER 1: INTRODUCTION

1.1 Project Background

In the modern digital retail landscape, e-commerce platforms facing intense market competition must deliver tailored consumer experiences to drive conversion and customer retention. The global cosmetics and luxury skincare sector presents unique customer discovery dynamics: consumers frequently exhibit complex skin profile requirements, brand loyalty patterns, and multi-step routine co-purchases. Rosevelle is a specialized e-commerce recommendation and retail analytics web platform engineered to satisfy both operational requirements.

1.2 Problem Statement & Objectives

Primary Objectives:

- **Multi-Model Recommendation Engine:** SVD, Collaborative Filtering, TF-IDF Content Filtering, Hybrid linear blending (RMSE = 1.0751).
- **Integrated Retail Analytics Suite:** Apriori Market Basket rules, RFM K-Means Customer Clustering, Multidimensional Sales OLAP Data Cubes.
- **Dual-Dataset Separation:** Real Amazon All Beauty ratings vs. Synthetic Cosmetics Order logs.

CHAPTER 4: DATASET SPECIFICATIONS

Table 4.1: Representative Real Amazon All Beauty Products Catalog

ASIN	Product Name	Brand	Price	Rating
B000143526	CeraVe Hydrating Facia...	CeraVe	■550.00	4.7
B0009V1YR8	The Ordinary Niacinami...	The Ordinary	■600.00	4.5
B001MA0QY2	Maybelline Lash Sensat...	Maybelline	■549.00	4.4
B003V21W3I	L'Oréal Paris Revitali...	L'Oréal Paris	■999.00	4.6
B004D248KC	Paula's Choice 2% BHA ...	Paula's Choice	■2,700.00	4.6
B00551GBWC	La Roche-Posay Antheli...	La Roche-Posay	■1,850.00	4.7
B007MT6J1E	NYX Soft Matte Lip Cre...	NYX	■699.00	4.4
B008630WNE	Neutrogena Hydro Boost...	Neutrogena	■950.00	4.6

CHAPTER 6 & 11: RECOMMENDATION & EVALUATION RESULTS

Table 11.1: Empirical Model Benchmark Performance

Model Architecture	RMSE	MAE	Configuration / Notes
Matrix Factorization (Truncated SVD)	1.0536	0.8477	4.0 Latent Factors
User-Based Collaborative Filtering	1.1289	0.9089	User Cosine Sim Matrix
Item-Based Collaborative Filtering	1.0966	0.8914	Item Cosine Sim Matrix
Content-Based Filtering (TF-IDF)	1.2041	0.9613	Metadata TF-IDF Vectors
Hybrid Model (SVD + Content TF-IDF)	1.0751	0.8739	Alpha = 0.6 Blend Weight

CHAPTER 13: SOFTWARE TESTING EXECUTION

Table 13.1: System Integration & Endpoint Test Matrix

ID	Test Module	Endpoint / Action	Result	Status
TC-01	Flask API Health Che	GET /api/health	200 OK, status: health	PASSED
TC-02	Dataset Explorer Met	GET /api/dataset-explo	JSON stats summary ret	PASSED
TC-03	Real Amazon Metadata	GET /api/dataset-recor	12 product metadata it	PASSED
TC-04	Real Review Records	GET /api/dataset-recor	85 review records	PASSED
TC-05	Synthetic Order Reco	GET /api/dataset-recor	108 order records	PASSED
TC-06	Hybrid Recommendatio	POST /api/recommend (u	Top 5 recommendations	PASSED

TC-07	Evaluation Metrics C	GET /api/evaluation-me	RMSE, MAE, Precision c	PASSED
TC-08	Apriori Rule Mining	GET /api/apriori-rules	14 association rules r	PASSED